

Continuous Assessment 1 (CA1): Usability Evaluation of PocketBook Reader

1. Evaluation Context

- **Application Name:** PocketBook Reader
- **Application Version:** v5.38 (Latest Stable Release)
- **Operating System:** Android 14 (Mobile OS)
- **Hardware Device:** Xiaomi Redmi Note 14 (Smartphone)
- **Evaluation Method:** Heuristic Evaluation based on **Jakob Nielsen's 10 Usability Heuristics for User Interface Design**.
- **Target Audience:** General readers seeking to manage, search, download, and customize digital e-books in multiple formats (EPUB and PDF).

2. Task Definition

Task Title: *Finding, Downloading, and Customizing a Book's Visual Theme*

To evaluate the application under realistic multi-step usage, the following sequence of consecutive actions was evaluated:

1. **Search & Filter:** Open the app, navigate to the internal Store/Catalog tab, search for a classic public-domain e-book (e.g., "*Housemaid*"), and apply a language/genre filter.
2. **Library Management:** Select the book, download/add it to the local library, and confirm its download status.
3. **Reader Execution:** Open the newly downloaded book from the local library.
4. **Display Customization:** Open the in-reader settings view, attempt to adjust the background color theme (from Light to Sepia or Dark Mode), change font size, and adjust margin widths.

3. Application of Nielsen's Usability Heuristics & Analysis

Three key heuristics were evaluated across this multi-step task flow:

Heuristic 1: Visibility of System Status (Jacob Nielsen's Heuristic 1st)

The design should always keep users informed about what is going on, through appropriate feedback within a reasonable time.

- **Evaluation: Partially Satisfied / Violated**
- **Observations & Interface Evidence:**
 - *Partial Success:* When searching and filtering books in the store, loading spinners don't clearly indicate network queries. Once inside the reader, page numbers (Page 12 of 340) update instantly as the user swipes.
 - *Usability Issue (Violation):* When tapping the download button for a book from the cloud or store catalog, the button briefly changes state, but there is no explicit progress bar or download percentage indicator showing the download speed or remaining time. If the file is large (e.g., a scanned PDF), the user is left wondering whether the download failed, is frozen, or is still actively transferring.
- **Proposed Improvement:**
 - Implement an active progress bar with a percentage counter directly over or beneath the book cover during downloading.
 - Add a transient toast notification (e.g., "Download complete — Added to Library") with a direct quick-action link to "Open Now".

Heuristic 2: Match Between System and the Real World (Jacob Nielsen's Heuristic 2nd)

The design should speak the users' language, with words, phrases, and concepts familiar to the user, rather than system-oriented terms.

- **Evaluation: Satisfied**
- **Observations & Interface Evidence:**
 - *Success:* PocketBook Reader excels at using real-world metaphors. The library navigation uses terms like "Bookshelf", "Favorites", and "Collections".
 - Inside the reader UI, turning pages uses a realistic 3D page-curl animation and physical swipe gestures that mimic flipping pages in a printed paper book. The brightness/font slider icons use standard, easily recognizable symbols (a sun icon for brightness, a small 'A' and large 'A' for text size scaling).

Heuristic 3: Recognition Rather Than Recall (Jacob Nielsen's Heuristic 6th)

Minimize the user's memory load by making objects, actions, and options visible. The user should not have to remember information from one part of the dialogue to another.

- **Evaluation: Violated**
- **Observations & Interface Evidence:**
 - *Usability Issue (Violation):* When adjusting font face, margins, and line spacing inside the reading menu, PocketBook Reader uses a slider menu overlaid on top of the text. However, many advanced controls (such as hyphenation, page turn animation types, or publisher defaults) are hidden deep within sub-menus labeled only with abstract icons lacking text labels.
 - Furthermore, when changing the background color/theme (e.g., switching from Day to Sepia mode), the settings windows obscures more than 50% of the active page text. Users cannot see the immediate visual effect of their typographic changes on a full paragraph of text without closing the menu wheel entirely, forcing them to rely on memory to assess whether the text size and contrast are comfortable.
- **Proposed Improvement:**
 - **Text Labels:** Add concise text labels beneath sub-menu icons inside the settings wheel.
 - **Non-Modal Bottom Sheet Preview:** Replace the centered radial wheel overlay with a compact bottom sheet that occupies only the lower third of the screen. This leaves the majority of the page visible so users can instantly preview font, margin, and color changes in real time before dismissing the controls.

Heuristic 4: Flexibility and Efficiency of Use (Jacob Nielsen's Heuristic 7th)

Accelerators — unseen by the novice user — may often speed up the interaction for the expert user such that the design can cater to both inexperienced and experienced users.

- **Evaluation: Satisfied**
- **Observations & Interface Evidence:**
 - *Success:* PocketBook Reader accommodates both casual and power readers seamlessly. Novice users can tap the edges of the screen or swipe to turn pages. Expert users can configure volume keys for page turns, set up two-finger vertical

drag gestures along the left edge to adjust brightness, and use pinch-to-zoom gestures to scale fonts on the fly without opening the settings menu at all.

4. Summary Table of Identified Issues & Proposed Improvements

Task Step	Identified Problem	Associated Heuristic	Proposed Solution
1. Download Book	Lack of precise progress feedback during book file downloading.	#1: Visibility of System Status	Display a dynamic percentage progress bar and a completion toast notice with a direct "Open" shortcut.
2. Reading Settings	Settings menu wheel covers half the page, blocking real-time font and theme preview.	#6: Recognition Rather Than Recall	Transition to a bottom-sheet control panel that leaves 70% of the page visible for live rendering.
3. Advanced Menu	Abstract icons without text labels force users to memorize or guess menu functions.	#6: Recognition Rather Than Recall	Append explicit textual tooltips/labels under all secondary configuration icons.